



# Banapuram Pranay Kumar

B.Tech Electrical Engineering

ee24btech11011@iith.ac.in

+919441174054



| Degree                        | University/Institute                         | Year | CGPA/Marks(%) |
|-------------------------------|----------------------------------------------|------|---------------|
| B.Tech Electrical Engineering | IIT Hyderabad                                | 2028 | 7.04          |
| XII (TGBIE)                   | BHAVAN'S SRI AUROBINDO JUNIOR COLLEGE        | 2024 | 98.80%        |
| X (CBSE)                      | ATOMIC ENERGY CENTRAL SCHOOL - 2 , HYDERABAD | 2022 | 95.20%        |

## POSITIONS OF RESPONSIBILITY

**Core Member**, Behind The Lens – Film Making Club

## SKILLS

**Programming languages** : Python , C , MATLAB , Verilog , LATEX

**Libraries and Frameworks** : NumPy , Pandas , Matplotlib , PyTorch , OpenCV , Scikit-Learn , Scipy , Librosa

**Machine Learning** : Supervised Learning , Deep Learning , Model Training and Evaluation

**Computer Vision** : Image processing , audio processing , Feature Extraction , Audio-Visual Learning

**Tools and Platforms** : Git , Github , Linux(Ubuntu) , Google Colab

## RELEVANT COURSES

**Mathematics** : Math for EE(Linear Algebra,Calculus) , Differential Equations and Transform Techniques, Probability,Random Processes,Matrix Theory

**Computer Science and AI** : Introduction to Programming(C) , Data Structures and Algorithms , Pattern Recognition and Machine Learning,Scientific Programming for EE

## PROJECTS

### Eigenvalue Computation using Hybrid Newton–Power Method

**Technologies:** C

- Designed and implemented a hybrid numerical algorithm combining the Power Method and Newton's Method for eigenvalue computation.
- Developed the complete implementation in C without using external numerical or linear algebra libraries.
- Implemented matrix operations, convergence criteria, and iterative optimization from first principles.
- Evaluated numerical accuracy and convergence characteristics on multiple test matrices.

### Synthetic Speech Attribution

**Technologies:** Python, PyTorch, NumPy

- Developed a deep learning model to identify the source of synthetic speech.
- Extracted and analyzed acoustic features from speech recordings.
- Trained and evaluated neural network models for speech classification.
- Investigated model performance across different synthetic speech generators.

### AVZoom: Audio-Visual Speaker Enhancement

**Technologies:** Python, PyTorch, OpenCV, NumPy

- Developed an audio-visual deep learning system to enhance the speech of a target speaker using synchronized video and audio inputs.
- Built data preprocessing pipelines for extracting and aligning visual and audio features.
- Implemented and trained convolutional neural network models for multimodal speech enhancement.
- Evaluated model performance using objective speech quality metrics.

### Speech Denoiser using Neural Networks

**Technologies:** Python, NumPy

- Designed and implemented a speech denoising system using feedforward neural networks developed entirely from scratch.
- Built the complete neural network pipeline, including forward propagation, backpropagation, gradient descent optimization, and parameter updates without using deep learning libraries.
- Trained the network to learn the mapping between noisy and clean speech signals.
- Evaluated denoising performance through quantitative loss analysis and qualitative listening tests.

### Machine Learning Algorithms from Scratch

**Technologies:** Python, NumPy, Matplotlib

- Implemented fundamental machine learning algorithms from first principles without relying on high-level machine learning libraries.
- Built complete training, prediction, and evaluation pipelines using Python and NumPy.
- Applied the implementations to multiple real-world problems, including:
  - Music vs Speech Classification
  - Heart Rate Prediction
  - Cricket Player Classifier
- Performed data preprocessing, feature engineering, visualization, and model evaluation.

## EXTRACURRICULAR

**Video Making** : Participated in film production, videography, and creative media projects.

**Public Speaking** : QuarterFinalist in Yuva-Sangarsh ( Inter Collegiate Leadership Competition)